

Rational Numbers on the Number Line

Lesson 9-4

Name: _____

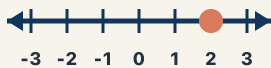
Date: _____

Class: _____

Key Vocabulary

Level 1 support

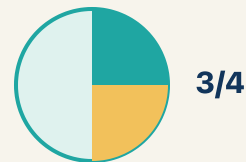
Picture first, then the word, then a plain-language meaning. Say each word out loud.



$\frac{1}{2} = 0.5$, $-\frac{3}{4} = -0.75$, $6 = \frac{6}{1}$ — all rational numbers

Rational number

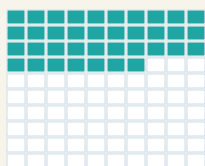
Any number you can write as a fraction. This includes fractions, decimals, and integers.



$\frac{3}{4}$ means 3 out of 4 equal parts — like 3 slices of a pizza cut into 4

Fraction

A number that shows part of a whole, like $\frac{3}{4}$.



$0.5 = \frac{1}{2}$ (halfway between 0 and 1 on the number line)

Decimal

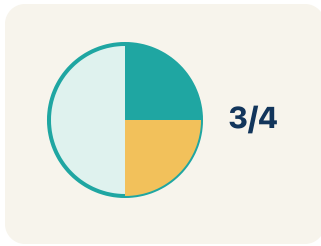
A number with a dot, like 0.5, that shows a part less than one.



A line with marks at -2, -1.5, -1, -0.5, 0, 0.5, 1, 1.5, 2

Number line

A straight line with numbers spaced out evenly.



$\frac{1}{2} = 0.5 = \frac{2}{4}$ — all name the same point on the number line

Equivalent

Having the same value, just written a different way.

...-2, -1, 0, 1, 2...



..., -3, -2, -1, 0, 1, 2, 3, ...

Integer

Whole numbers and their opposites, like -2, -1, 0, 1, 2.

Key Ideas & Notes



WHAT WE'RE LEARNING TODAY

I can place rational numbers, including fractions and decimals, on a number line.

1 Notice

Captain Vega's advanced treasure map uses precise coordinates — not just whole numbers!

2 Key idea

I can place rational numbers, including fractions and decimals, on a number line.

3 Words I'll use

Rational number

Fraction

Decimal

Number line

Equivalent

Integer



Think About It

- Where would -2.5 fall on a number line between -3 and -2 ?
- How is plotting $3/4$ different from plotting a whole number?
- Can a negative number be a fraction or decimal?

See the notes in action: watch one worked all the way through, then try the next with the same steps.



I do — watch

Follow each step as your teacher solves it.

Problem: Which point is located between -1 and -2 on the number line?

- A. -0.5
- B. -1.5
- C. -2.5
- D. 1.5

Step 1

-1.5 is between -1 and -2 because it is 0.5 units to the left of -1 (or 0.5 units to the right of -2).



Answer: B. -1.5

 We do — together

Solve this one with your class using the same steps.

Problem: Which fraction is equivalent to 0.25?

- A. $\frac{1}{4}$
- B. $\frac{1}{2}$
- C. $\frac{2}{5}$
- D. $\frac{1}{3}$

Step 1

Step 2

Answer:

 You do — your turn

Now try one on your own. Show every step.

Problem: Where is $-\frac{3}{4}$ located on the number line?

- A. Between 0 and -1
- B. Between -3 and -4
- C. Between -1 and -2
- D. Between 0 and 1

Show your work:

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

How can a fraction like $\frac{1}{2}$, a decimal like 0.75, and an integer like -3 all be rational numbers?

Level 1 support

Start here: All of these are rational numbers because each one can be ____.

 **Need a hint?**

Sentence starters (optional)

A rational number is any number you can write as ____.

Un numero racional es cualquier numero que puedes escribir como ____.

For example, -3 can be written as ____.

Por ejemplo, -3 se puede escribir como ____.

Word bank: rational number fraction decimal integer number line

Listen for: Student explains a rational number is any number that can be written as a fraction, including integers and decimals.

Level 2

Push students to explain how to place $\frac{1}{2}$, 0.75, and -3 on the same number line.

- To place 0.75 on the number line, I would ____.
- I can compare a fraction and a decimal by ____.

EXPLORE

Where would you place $-1\frac{1}{2}$ on the number line, and how do you decide between which two integers it falls?

Level 1 support

Start here: $-1\frac{1}{2}$ falls between ____ and ____ because ____.

 **Need a hint?**

Sentence starters (optional)

$-1\frac{1}{2}$ is between ____ and ____.

$-1\frac{1}{2}$ esta entre ____ y ____.

It is closer to ____ because ____.

Esta mas cerca de ____ porque ____.

Word bank: rational number fraction number line integer negative

Listen for: Student places $-1\frac{1}{2}$ between -1 and -2 , halfway, reasoning from the value of the fraction part.

Level 2

Ask students to compare $-1\frac{1}{2}$ and -1 using the number line and justify which is greater.

- -1 is greater than $-1\frac{1}{2}$ because ____.
- On the number line, ____ is to the right of ____.

CONNECT

How can rewriting a fraction as a decimal (or a decimal as a fraction) help you compare two rational numbers?

Level 1 support

Start here: Rewriting the numbers in the same form helps because ____.



Need a hint?

Sentence starters (optional)

If I turn $\frac{3}{4}$ into a decimal, I get ____.

Si convierto $\frac{3}{4}$ en decimal, obtengo ____.

Now I can compare them because ____.

Ahora puedo compararlos porque ____.

Word bank: rational number fraction decimal compare number line

Listen for: Student explains converting to a common form (both decimals or both fractions) makes comparison and ordering easier.

Level 2

Push students to generalize a strategy for ordering a mixed list of fractions, decimals, and integers.

- My strategy for ordering a mixed list is ____.
- Putting them all in the same form helps because ____.

REFLECT

Is every integer a rational number? Is every rational number an integer? Explain your thinking.

Level 1 support

Start here: Every integer is a rational number, but ____.



Need a hint?

Sentence starters (optional)

Every integer is rational because it can be written as ____.

Todo entero es racional porque se puede escribir como ____.

But not every rational number is an integer because ____.

Pero no todo numero racional es un entero porque ____.

Word bank: rational number integer fraction decimal number line

Listen for: Student explains integers are rational (write as a fraction over 1) but rationals like $\frac{1}{2}$ are not integers.

Level 2

Ask students to critique the statement 'all fractions are integers' with an example.

- The statement is false because ____.
- An example that disproves it is ____.

Write About the Math

The Writing Revolution

I can explain my reasoning using the words rational number, fraction, decimal, and number line.

1. Kernel Sentence subject + verb

Model: Rational number is any number you can write as a fraction. This includes fractions, decimals, and integers.

Número racional es cualquier número que puedes escribir como fracción. Incluye fracciones, decimales y enteros.

Write a kernel sentence about rational number. Use a subject and a verb.

Escribe una oración base sobre número racional. Usa un sujeto y un verbo.

2. Sentence Expansion because • but • so

Kernel: Rational number matters in math

Número racional importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Rational number matters in math because ____.

Número racional importa en matemáticas porque ____.

but
pero

Rational number matters in math, but ____.

Número racional importa en matemáticas, pero ____.

so
entonces

Rational number matters in math, so ____.

Número racional importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement Afirmación

Tell one true fact about rational number.
Di un hecho verdadero sobre rational number.

Rational number ____.

Question Pregunta

Ask a question about rational number.
Haz una pregunta sobre rational number.

How does ____ ?

¿Cómo ____ ?

Exclamation Exclamación

Show excitement about rational number.
Muestra entusiasmo sobre rational number.

Wow, ____ !

¡Guau, ____ !

Command Mandato

Tell a partner what to do with rational number.
Dile a un compañero qué hacer con rational number.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

A rational number is any number you can write as ____.

Un numero racional es cualquier numero que puedes escribir como ____.

For example, -3 can be written as ____.

Por ejemplo, -3 se puede escribir como ____.

-1 1/2 is between ____ **and** ____.

-1 1/2 esta entre ____ *y* ____.

Try It

Solve on your own. Check the answer key when you are done.

1. What is the opposite of $-\frac{2}{3}$?

- A. $\frac{2}{3}$
- B. $-\frac{3}{2}$
- C. $\frac{3}{2}$
- D. $-\frac{2}{3}$

Show your work:

2. Order from least to greatest: -0.5 , 0.25 , -1 , 0.75 .

- A. -1 , -0.5 , 0.25 , 0.75
- B. -0.5 , -1 , 0.25 , 0.75
- C. 0.75 , 0.25 , -0.5 , -1
- D. -1 , 0.25 , -0.5 , 0.75

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Name three different rational numbers between -1 and 0. Write each as both a fraction and a decimal. Explain how you know they are between -1 and 0.

Sentence starter: Three rational numbers between -1 and 0 are: $\frac{\quad}{\quad} = \quad$, $\frac{\quad}{\quad} = \quad$, and $\frac{\quad}{\quad} = \quad$. I know they are between -1 and 0 because $\quad$.

Show your work:

Reflect — Exit Ticket

Which rational number is located between -2 and -3 on the number line?

- A. -1.5
- B. -2.75
- C. -3.5
- D. 2.5

Your answer:

Answer Key & Teacher Guide

1. **We Try (notes):** A. $1/4$ — 0.25 means 25 hundredths = $25/100 = 1/4$.
2. **You Try (notes):** A. Between 0 and -1 — $-3/4 = -0.75$, which is between 0 and -1. It is $3/4$ of the way from 0 toward -1.
3. **Try It 1:** A. $2/3$ — The opposite of $-2/3$ is $2/3$, the same distance on the other side of 0.
4. **Try It 2:** A. -1, -0.5, 0.25, 0.75 — Most negative first: -1, -0.5, 0.25, 0.75.
5. **Exit Ticket:** B. -2.75 — -2.75 is between -2 and -3 because it is 0.75 units to the left of -2. -1.5 is between -1 and -2, and -3.5 is between -3 and -4.

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Rational number is any number you can write as a fraction. This includes fractions, decimals, and integers.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).